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Tackling decision dependency in contextual stochastic optimization

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Abstract

In this paper, we study contextual stochastic optimization, where decisions are made under uncertainty and the distribution of random parameters can be partially inferred from covariates observed prior to decision-making. In many practical settings, these distributions also depend on the decisions themselves, a phenomenon known as the *decision-dependent effect*. Most existing studies address this issue by imposing structural assumptions on the relationship between decisions and the underlying distributions. However, such assumptions may lead to model misspecification when the true relationship deviates from the assumed form. A prominent alternative is the weighted sample average approximation (wSAA) method, which adapts sample weights based on their similarity to the current decision–context pair. Nevertheless, because these weights are typically computed using complex machine learning models and depend on the decision variables in decision-dependent settings, solving the resulting optimization problem becomes computationally challenging. To overcome this challenge, we extend the wSAA framework from the loss function to its gradient, leading to the notion of the *contextual gradient*. We show that the contextual gradient serves as a meaningful indicator of optimality and leverage this property to develop the *contextual gradient descent (CGD)* algorithm. Our analysis establishes that CGD converges to a neighborhood of the global optimum when the loss function exhibits sufficient strong convexity. Moreover, the derived bounds reveal a key insight: the strength of convexity in the loss function can compensate for the uncertainty introduced by decision-dependent effects. Extensive numerical experiments on both synthetic and real-world datasets demonstrate that CGD consistently outperforms existing methods for contextual optimization under decision-dependent uncertainty.

Keywords

Contextual Optimization, Prescriptive Analytics, Decision-Dependency, Weighted Sample Average Approximation, Gradient Descent

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1 Introduction

In many real-world management processes, decision-makers must make decisions under uncertainty. These problems are often formulated as stochastic optimization models of the form:

$$\min_{x\in\mathcal{X}} g(\boldsymbol{x}) = \mathbb{E}_{\boldsymbol{y}\sim f(\boldsymbol{y})}[l(\boldsymbol{x},\boldsymbol{y})], \tag{1}$$

where $\boldsymbol{x}$ is the decision variable within the feasible region $\mathcal{X} \subseteq \mathbb{R}^d$; $\boldsymbol{y}$ is a random parameter with a probability density function $f(\boldsymbol{y})$; $l(\boldsymbol{x},\boldsymbol{y})$ is the loss function, which evaluates the performance of decision $\boldsymbol{x}$ under a realization of $\boldsymbol{y}$; and $g(\boldsymbol{x})$ is the expectation of $l(\boldsymbol{x},\boldsymbol{y})$ with respect to $f(\boldsymbol{y})$.

When the distribution of $\boldsymbol{y}$ is influenced by covariates $\boldsymbol{z}$, which are observed before decision-making, it becomes beneficial to make the decision condition on $\boldsymbol{z}$ to improve decision quality. This leads to the framework of *contextual stochastic optimization (CSO)*. The CSO paradigm has recently

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received growing attention in the operations research community. A comprehensive review of related work can be found in Sadana et al. (2025). Under the CSO setting, the stochastic optimization problem is reformulated as:

$$\min_{\boldsymbol{x}\in\mathcal{X}} g(\boldsymbol{x},z) = \mathbb{E}_{\boldsymbol{y}\sim f(\boldsymbol{y}|z)}[l(\boldsymbol{x},\boldsymbol{y})]. \tag{2}$$

Moreover, in many applications, the stochastic parameters themselves may also depend on the decisions. A typical example is the pricing problem, where a firm must determine the optimal price for a product before observing actual demand. In this case, demand is directly influenced by the price—higher prices generally lead to lower demand, and vice versa. This leads to a decision-dependent CSO problem formulated as

$$\min_{\boldsymbol{x}\in\mathcal{X}} g(\boldsymbol{x},z) = \mathbb{E}_{\boldsymbol{y}\sim f(\boldsymbol{y}|\boldsymbol{x},z)}[l(\boldsymbol{x},\boldsymbol{y})]. \tag{3}$$

A key challenge in solving decision-dependent CSO problems is that the conditional distribution $f(\boldsymbol{y} \mid \boldsymbol{x}, z)$ is rarely known to the decision-maker, and there is often insufficient data to estimate it accurately because samples are sparse across different contexts and decision values. Prior studies typically address this issue by imposing structural assumptions on the conditional distribution. For example, Chu et al. (2025) studied the price-setting newsvendor problem under the assumption that the expected demand follows a linear or log-linear function of price.

However, such structural assumptions can lead to model misspecification when the true decision-dependent effect deviates from the assumed form. A promising nonparametric approach for incorporating decision dependence in CSO is the weighted sample average approximation (wSAA) method proposed by Bertsimas and Kallus (2019). This method enhances the classical sample average approximation (SAA) framework by assigning a weight to each sample that reflects its relevance to the current decision–context pair, thereby providing a flexible data-driven estimate of the objective function in problem (3). Specifically, the wSAA approach is formulated as

$$\min_{\boldsymbol{x}} \ \hat{g}(\boldsymbol{x},z) \overset{\Delta}{=} \sum_{i=1}^{N} w^{(i)}(\boldsymbol{x},z)\, l(\boldsymbol{x},\boldsymbol{y}^i), \tag{4}$$

where $w^{(i)}(\boldsymbol{x},z)$ is a weight function that specifies the relevance of the i-th historical data point to the evaluated decision–context pair $(\boldsymbol{x},z)$, and $l(\boldsymbol{x},\boldsymbol{y}^i)$ represents the loss incurred when the decision $\boldsymbol{x}$ is applied and the realized random outcome is given by the i-th sample $\boldsymbol{y}^i$. The core idea is to assign higher weights to historical samples that are more similar to the current decision–context pair, thereby facilitating more accurate estimation of the conditional distribution. In practice, these weights are typically computed using machine learning (ML) models before solving problem (4). Representative weighting schemes are introduced below, and more detailed discussions can be found in Bertsimas and Kallus (2019) and Lin et al. (2022).

Example 1 (k-nearest neighbor (kNN) weight)**.** *The weight function based on the kNN method is defined as:*

$$w^{kNN,(i)}(\boldsymbol{x},z) = \frac{1}{k}\mathbb{I}\{(\boldsymbol{x}^i,z^i) \textit{ is among the kNNs of } (\boldsymbol{x},z)\}, \quad \forall i \in [N], \tag{5}$$

where $\mathbb{I}\{\cdot\}$ is the indicator function.

Example 2 (Kernel regression (KR) weight)**.** *Weights can also be computed via kernel functions based on the distance between $(\boldsymbol{x},z)$ and $(\boldsymbol{x}^i,z^i)$:*

$$w^{KR,(i)}(\boldsymbol{x},z) = \frac{K_h\left(\|(\boldsymbol{x},z)-(\boldsymbol{x}^i,z^i)\|_2\right)}{\sum_{j=1}^{N} K_h\left(\|(\boldsymbol{x},z)-(\boldsymbol{x}^j,z^j)\|_2\right)}, \tag{6}$$

where $K_h : \mathbb{R}^{\dim(\boldsymbol{x})+\dim(z)} \to \mathbb{R}$ is a kernel function with bandwidth h. Common choices include uniform, triangular, and Gaussian kernels. Unless otherwise noted, we adopt the Gaussian kernel:

$$K_h(z) = C_K \exp\left(-\frac{\|z\|_2^2}{2}\right)\mathbb{I}\left[\|z\|_2 \le h\right], \tag{7}$$

where $C_K = 1/\int_{z\in\mathcal{Z}} \exp\left(-\|z\|_2^2/2\right)\mathbb{I}[\|z\|_2 \le h]dz$ is a normalization constant.

Example 3 (Classification and regression tree (CART) weight)**.** *CART-based weights are defined using leaf membership in decision trees:*

$$w^{CART,(i)}(\boldsymbol{x},z) = \frac{\mathbb{I}\{R(\boldsymbol{x},z)=R(\boldsymbol{x}^i,z^i)\}}{|\{j : R(\boldsymbol{x}^j,z^j)=R(\boldsymbol{x},z)\}|}, \tag{8}$$

where $R : \mathcal{X}\times\mathcal{Z} \to \{1,\dots,r\}$ maps each sample to one of the r terminal leaves in the regression tree.

Example 4 (Random forest (RF) weight)**.** *RF weights average across multiple CART estimators:*

$$w^{RF,(i)}(\boldsymbol{x},z) = \frac{1}{N_E}\sum_{e=1}^{N_E} w^{CART,i,e}(\boldsymbol{x},z), \tag{9}$$

where N_E is the number of estimators, and $w^{CART,i,e}$ is the weight assigned by the e-th tree.

Although the wSAA method achieves asymptotic optimality under certain conditions (see Theorem 10 in Bertsimas and Kallus, 2019), the resulting optimization problem is difficult to solve. Because the decision variables appear in the weights

determined by complex ML methods, the estimated objective function becomes discrete or nonconvex with respect to the decision variables—even when $l(\boldsymbol{x}, \boldsymbol{y})$ is continuous and convex. To preserve tractability, Bertsimas and Kallus (2019) proposed a discretization strategy and developed a tailored solution approach for the tree-based weight case. However, as we show in Section 5.3.1, the method incurs a significant computational burden.

This research aims to develop a more efficient approach for solving decision-dependent CSO problems under the wSAA framework. Instead of applying the wSAA method directly to the loss function, we apply it to the gradient of the loss function, leading to what we call the *contextual gradient*. Unlike the gradient of the original wSAA objective, the contextual gradient does not require computing the derivative of the sample weights, which are typically obtained from complex ML models. Under mild regularity conditions and for specific weighting schemes, we show that the contextual gradient provides a reliable measure of optimality in decision-dependent CSO problems and can therefore serve as an effective search direction within a gradient descent (GD) framework. Leveraging this property, we propose the *contextual GD (CGD)* algorithm. Due to this computational simplification, the CGD algorithm may not converge to the global optimum in general settings. We establish a sufficient condition under which the algorithm converges to a neighborhood of the optimal solution. In particular, when the loss function exhibits sufficient strong convexity relative to the scale of the decision-dependent effect, CGD guarantees a bounded optimality gap with respect to the global optimum, where the bound depends on both the strength of convexity and the magnitude of the decision-dependent effect. Extensive numerical experiments on both synthetic and real-world datasets demonstrate that our method consistently outperforms existing approaches in terms of both solution quality and computational efficiency.

1.1 Contributions

Our contributions are threefold.

First, we propose an efficient and effective nonparametric approach for solving CSO problems with decision-dependent effects. We introduce the concept of the *contextual gradient* and show that, under mild conditions, it serves as an asymptotically unbiased estimate of the expected gradient of the loss function and remains bounded at the optimal solution, with its magnitude determined by the degree of decision dependence. Building on this property, we develop the CGD algorithm, which incorporates the contextual gradient into a standard GD framework. Under mild assumptions, we establish convergence guarantees: when the loss function exhibits sufficient strong convexity relative to the scale of the decision-dependent effect, CGD converges to a neighborhood of the global optimum, with the neighborhood size determined by the relative magnitude of the decision-dependent effect and the strength of convexity. Extensive numerical experiments demonstrate that, compared with existing wSAA-based solution approaches, CGD achieves comparable solution quality with substantially lower computational time.

Second, our analysis reveals how strong convexity compensates for decision-dependent effects. In particular, under strong convexity, the optimality gap is shown to scale with the ratio between the degree of decision dependence and the strength of convexity. Since the contextual gradient is constructed by omitting part of the derivative of the wSAA objective, this result implies that strong convexity mitigates the information loss introduced by this simplification. In other words, even when part of the decision-dependent structure is ignored during optimization, the algorithm can still achieve desirable performance provided that the loss function exhibits sufficient curvature.

Third, we validate CGD through synthetic and real-world experiments. Compared with the wSAA approach of Bertsimas and Kallus (2019), CGD achieves lower optimality gaps with substantially reduced computational time. Furthermore, when compared with parametric estimate-then-optimize (ETO), smart predict-then-optimize (SPO), and end-to-end approaches, CGD performs competitively when the data follow the assumed parametric model and significantly outperforms these methods when the data-generating process deviates from the assumed structure. In a real-world case study, we further demonstrate the practical benefits of the proposed method.

1.2 Examples of decision-dependent CSO problem

Our proposed CGD algorithm is designed for CSO problems with decision-dependent effects and continuous decision variables. Such settings are common in practical applications. To provide clear motivation for our study, we illustrate three representative examples.

Example 5 (Price-setting newsvendor problem)**.** *A decision-maker repeatedly sells a perishable product under demand uncertainty. In each period, she jointly chooses a selling price p and an order quantity q, denoted by $\boldsymbol{x} = (p, q)^T$, to maximize expected revenue. Let c be the unit cost and s the salvage value. For realized demand y, the loss function is*

$$l(\boldsymbol{x}, y) = -(p - c)q + (p - s)(q - d)^+.$$

As documented in prior studies (Ban and Rudin, 2018; Chu et al., 2025), demand depends on both observable covariates $\boldsymbol{z}$ (e.g., demographics, weather, and seasonality) and the price p, yielding a decision-dependent CSO problem of the form (3).

Example 6 (Pop-up store placement)**.** *Location optimization on a continuous map is a classical problem in operations research (Liu et al., 2025). Consider a retailer operating pop-up stores: at the beginning of each day, the retailer selects a*

deployment location $\boldsymbol{x} \in \mathbb{R}^2$ within an urban region $\mathcal{X}$ (Cao et al., 2021). Assuming a fixed unit price p and that all realized demand is served, the loss function for demand realization y is

$$l(\boldsymbol{x}, y) = -p\, y.$$

Although the location $\boldsymbol{x}$ does not explicitly enter the loss function, demand depends on it through factors such as visibility, accessibility, and foot traffic. In addition, contextual variables $\boldsymbol{z}$ (e.g., time of day, weather, local events, and neighborhood demographics) further affect demand. Consequently, the problem takes the form of a decision-dependent CSO problem (3).

Example 7 (Contextual project portfolio optimization with endogenous returns)**.** *Consider a case where a firm allocates capital across a portfolio of d projects. Let $\boldsymbol{x} \in \mathbb{R}^d$ denote the vector of investment proportions, satisfying $\sum_{i=1}^{d} x_i = 1$ and $x_i \geq 0$. Let $\boldsymbol{y} \in \mathbb{R}^d$ be the random vector of project returns. A common objective is to maximize expected return while controlling downside risk, which can be measured by the conditional value-at-risk at confidence level α. As in Rockafellar and Uryasev (2000), the corresponding loss function can be formulated as*

$$l((\boldsymbol{x}, \eta), \boldsymbol{y}) = -\boldsymbol{x}^\top \boldsymbol{y} + \rho\Big(\eta + \frac{1}{1-\alpha}(-\boldsymbol{x}^\top \boldsymbol{y} - \eta)^+\Big),$$

where $\rho > 0$ is a risk-aversion parameter, η is an auxiliary variable, and $a^+ = \max\{a, 0\}$.

Project returns depend on both contextual variables $\boldsymbol{z}$ (e.g., macroeconomic conditions, market sentiment, and industry forecasts) and the investment decision $\boldsymbol{x}$ itself. In particular, allocating excessive capital to similar or competing projects may induce internal competition and reduce marginal returns. Consequently, the problem is a decision-dependent CSO problem of the form (3).

1.3 Organizations and notations

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 presents the concept of the contextual gradient and analyzes its statistical properties. The proposed CGD algorithm and its convergence properties are presented in Section 4. In Section 5, we conduct extensive numerical experiments to validate the performance of the CGD algorithm and compare it with existing methods using synthetic data. A case study based on real-world data is provided in Section 6. Section 7 concludes the paper. All technical proofs are deferred to the E-Companion (EC) for clarity.

For simplicity of notation, we use $x \wedge y$ to denote $\min\{x, y\}$ and $(x)^+$ to denote $\max\{x, 0\}$. We let ∇ denote the gradient and ∂ denote the subgradient set. Let $\boldsymbol{x} = (x_1, \ldots, x_d)^T$. Similarly, all the vectors are presented in bold. The subscript denotes the corresponding coordinate of a vector, and $\mathbf{1}$ denotes the all-one vector. We use $\|\cdot\|$ to denote the ℓ_2 norm for vectors and matrices. A function f is L-Lipschitz continuous on $\boldsymbol{x} \in \mathcal{X}$ if $\|f(\boldsymbol{x}_1) - f(\boldsymbol{x}_2)\| \leq L\|\boldsymbol{x}_1 - \boldsymbol{x}_2\|$ for any $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathcal{X}$. A function f is γ-strongly convex if $f(\boldsymbol{x}_1) - f(\boldsymbol{x}_2) - \nabla f(\boldsymbol{x}_2)^T(\boldsymbol{x}_1 - \boldsymbol{x}_2) \geq (\gamma/2)\|\boldsymbol{x}_1 - \boldsymbol{x}_2\|^2$. We use $[N] := \{1, \ldots, N\}$ to denote the subscript set.

2 Literature review

CSO has emerged as a focal topic in the operations research community in recent years, and multiple frameworks have been proposed to address this problem. A widely adopted framework is the ETO paradigm, which first estimates the conditional distribution of the uncertain parameters given contextual covariates and then optimizes the decision based on this estimate. Our work is closely related to this stream of literature. Among existing approaches, the most relevant is the wSAA framework proposed by Bertsimas and Kallus (2019), which estimates the conditional distribution via a weighted empirical distribution. Beyond wSAA, residual-based approaches (Ban et al., 2019; Kannan et al., 2025) constitute another major class of ETO methods. These methods employ ML models, such as linear regression or decision trees, to capture the relationship between covariates and uncertain parameters, and then use the residuals to construct uncertainty sets for downstream optimization. Another influential line of work is the SPO framework proposed by Elmachtoub and Grigas (2022), which incorporates decision loss directly into the estimation procedure. While decision dependence can, in principle, be accommodated in these approaches by including decision variables in the predictive model, doing so typically introduces significant computational challenges and remains difficult to handle in a tractable manner. In contrast to ETO-based methods, end-to-end learning approaches directly learn a decision policy from data, bypassing an explicit prediction stage. For instance, Ban and Rudin (2018) proposed a linear decision rule for the newsvendor problem and optimized its parameters using training data, demonstrating improved performance over standard SAA. However, this line of work (e.g., Zhang and Gao, 2017) typically assumes that uncertainty is independent of the decision variables, and thus cannot capture decision-dependent effects in a principled and tractable manner. More recently, Feng and Shanthikumar (2023) proposed the operational data analysis (ODA) framework, which unifies ETO and end-to-end approaches by introducing a data integration model that projects raw data onto the decision space. Due to its generality, our proposed CGD algorithm can be viewed as a special case of the ODA framework, where minimizing the contextual gradient serves as the validating model. While Chu et al. (2025) applied ODA to price-setting newsvendor problems and incorporated decision dependence, their approach relies on strong structural assumptions on the price–demand relationship. Compared to the above-mentioned literature, our work contributes to the decision-dependent CSO literature

by providing a more flexible and computationally tractable solution framework without imposing restrictive parametric assumptions.

Decision-dependent uncertainty has also received increasing attention in the broader stochastic programming (SP) community without considering covariates. Dupacová (2006) highlighted the inherent computational challenges of such models and emphasized that tractability often relies on problem-specific structural assumptions. More recent work has aimed to develop more general solution approaches. For instance, Mendler-Dünner et al. (2020) proposed an iterative GD method for decision-dependent SP, and Liu et al. (2021) introduced a coupled learning-enabled optimization algorithm, which estimates the conditional distribution via local linear regression within a carefully designed trust region. Apart from not explicitly incorporating contextual information into the decision-making process, this line of literature differs from our work in that it assumes online access to an oracle capable of sampling from the true distribution under any given decision. In contrast, our method is designed for an offline setting, where only a fixed historical dataset is available, and no additional samples can be collected for alternative decisions.

In addition, from an algorithmic perspective, our work is related to the literature on GD methods with biased gradient estimators. When exact gradients are computationally expensive or infeasible to obtain, a large body of work has studied the convergence of algorithms that relied on biased gradient surrogates or gradient-free approximations. These studies were typically characterized by assumptions about bias. For instance, d'Aspremont (2008) showed that GD retained its optimal convergence rate when the gradient error was uniformly bounded. Devolder et al. (2014) developed an inexact oracle framework that incorporated bias in both the objective and its gradient, and analyzed the convergence of standard and accelerated methods. Chen and Luss (2018) considered biased gradient estimators arising from finite samples and showed that, under consistency conditions, convergence rates comparable to those of unbiased *stochastic GD (SGD)* could be achieved across strongly convex, convex, and nonconvex settings. Hu et al. (2021) further characterized convergence via linear matrix inequalities that related additive and multiplicative errors to properties of the loss function. In the context of CSO, Hu et al. (2020) studied biased SGD where the bias arose from sampling-based approximations of the true gradient. For a comprehensive review, we referred readers to Demidovich et al. (2023). Compared to the literature, the bias in our setting arose from systematically neglecting part of the decision-dependent effect when constructing the contextual gradient, rather than from stochastic noise or computational approximations. As a result, the bias was structural and problem-intrinsic. This distinction enabled us to explicitly characterize how the interplay between strong convexity and decision-dependent effects governed the resulting performance guarantees, going beyond a direct application of existing inexact-gradient theory.

3 Contextual gradient and its statistical properties

Despite its flexibility, solving the wSAA problem (4) poses significant computational challenges due to the dependence of the weights on the decision variable $\boldsymbol{x}$. Specifically:

- **Nonconvexity:** Since $\boldsymbol{x}$ appears in both the weights and the loss function, interaction (cross-product) terms arise, which may render the overall objective nonconvex—even when $l(\boldsymbol{x}, \boldsymbol{y})$ is convex in $\boldsymbol{x}$.
- **Nonsmoothness:** When ML models such as kNN or CART are used to compute the weights, the resulting objective function can be nonsmooth or even discontinuous, making standard gradient-based methods inapplicable.

Although Bertsimas and Kallus (2019) proposed a discretization-based solution, we show in Section 5.3.1 that it is computationally expensive to implement at scale. The above difficulties also make it challenging to apply gradient-based methods to solve the problem. Once the weights $w^{(i)}(\boldsymbol{x}, z)$ are specified by an ML model, computing the full gradient of the estimated objective (4) requires evaluating

$$\nabla_{\boldsymbol{x}} \hat{g}(\boldsymbol{x}, z) = \sum_{i=1}^{N} \left[w^{(i)}(\boldsymbol{x}, z)\, \nabla_{\boldsymbol{x}} l(\boldsymbol{x}, \boldsymbol{y}^i) + \nabla_{\boldsymbol{x}} w^{(i)}(\boldsymbol{x}, z)\, l(\boldsymbol{x}, \boldsymbol{y}^i) \right].$$

The main difficulty of computing the full gradient lies in evaluating $\nabla_{\boldsymbol{x}} w^{(i)}(\boldsymbol{x}, z)$. For kernel-based weights, this term involves analytically complex exponential expressions, while for kNN or CART weights, the gradients do not even exist.

3.1 Contextual gradient

To avoid the computation of the gradients of the weights, we apply the idea of wSAA directly to the gradient of the loss function to obtain what we define as *contextual gradient*, which is formally presented in Definition 1.

Definition 1 (Contextual gradient)**.** G*iven a contextual variable* $\boldsymbol{z}$ *and a decision* $\boldsymbol{x}$*, the contextual gradient at* $\boldsymbol{x}$ *based on the dataset* $\{(\boldsymbol{z}^i, \boldsymbol{x}^i, \boldsymbol{y}^i)\}_{i=1}^N$ *is defined as:*

$$\hat{G}_N(\boldsymbol{x}; z) = \sum_{i=1}^{N} w^{(i)}(\boldsymbol{x}, z)\, \nabla_{\boldsymbol{x}} l(\boldsymbol{x}, \boldsymbol{y}^i), \tag{10}$$

where $w^{(i)}(\cdot, \cdot)$ *is the weight function obtained from historical data.*

The weights used in the contextual gradient (10) are derived using the same procedure as in the wSAA (4). Compared to directly computing the gradient of the estimation of the objective (4), the contextual gradient avoids taking derivatives of the weights, which circumvents several computational challenges mentioned in the last section.

In many applications, computing the gradient of the loss function may be more challenging than evaluating the loss itself. However, in our setting, computing the gradient with respect to any decision $\boldsymbol{x}$ requires no stronger assumption than evaluating the loss for arbitrary decision–outcome pairs $(\boldsymbol{x}, \boldsymbol{y})$, as in the wSAA approach (4). Both rely on the assumption that $l(\boldsymbol{x}, \boldsymbol{y})$ has a known closed-form expression. Such evaluability is realistic in practice, since loss functions in operations settings are typically determined by cost and revenue structures known to the decision-maker, and all the illustrative examples in Section 1.2 satisfy this condition. For instance, in the price-setting newsvendor problem (Example 5), with $\boldsymbol{x} = (p, q)$, the subgradient of the loss function can be easily computed as

$$\partial_{\boldsymbol{x}} l(\boldsymbol{x}, y) = \left\{ \begin{bmatrix} -(y \wedge q) \\ -(p-c) + (p-s)e \end{bmatrix} : e \in \left[\mathbb{I}\{q > y\}, \mathbb{I}\{q \geq y\}\right] \right\}.$$

Moreover, it is easy to observe that the contextual gradient is part of $\nabla_{\boldsymbol{x}} \hat{g}(\boldsymbol{x}, \boldsymbol{z})$ and the component $\nabla_{\boldsymbol{x}} w^{(i)}(\boldsymbol{x}, \boldsymbol{z})\, l(\boldsymbol{x}, \boldsymbol{y}^i)$ is neglected. Although this simplification avoids the difficulty of computing the gradient of the weights, an important question arises: what is the statistical meaning of the contextual gradient, and how is it related to optimality?

3.2 Assumptions

To facilitate the subsequent theoretical analysis, we impose the following assumptions.

Assumption 1 (Bounded and uniform support)**.** *The support of the random parameter $\boldsymbol{y}$ is bounded and remains consistent across all decisions $\boldsymbol{x} \in \mathcal{X}$, where $\mathcal{X}$ is bounded, compact, and convex.*

Assumption 1 is commonly satisfied in practice, as one can define a unified support by taking the union of all feasible domains of $\boldsymbol{y}$ over the decision space $\mathcal{X}$, and assign zero probability to regions outside the domain of $f(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{z})$. We denote this common support of y as $\Omega \subseteq \mathbb{R}^q$, and its Lebesgue measure (volume) as S_Ω.

Assumption 2 (ε-sensitivity)**.** *The conditional density function $f(\boldsymbol{y} \mid \cdot, \boldsymbol{z})$ is ε-sensitive with respect to the decision variable. That is, for all $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathcal{X}$,*

$$W_1\big(f(\boldsymbol{y} \mid \boldsymbol{x}_1, \boldsymbol{z}), f(\boldsymbol{y} \mid \boldsymbol{x}_2, \boldsymbol{z})\big) \leq \varepsilon \|\boldsymbol{x}_1 - \boldsymbol{x}_2\|, \tag{11}$$

where W_1 denotes the Earth Mover's (Wasserstein-1) distance (Rubner et al., 2000).

Assumption 2 limits the extent to which the conditional distribution $f(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{z})$ can change as the decision $\boldsymbol{x}$ varies. We refer to ε as the *decision-dependence coefficient*, which quantifies the sensitivity of the distribution to changes in the decision variable. A larger ε indicates stronger decision-dependent effects. This condition holds for a broad class of models. For example, as discussed in Perdomo et al. (2020), Gaussian distributions with decision-dependent mean and variance satisfy this assumption. Assumption 2 is also satisfied by a regression-type model, that is, the decision only influences the mean and the mean is Lipschitz continuous in $\boldsymbol{x}$.

Assumption 3 (Smooth density function)**.** *The probability density function $f(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{z})$ has a L_g-Lipschitz-continuous gradient with respect to the decision variable $\boldsymbol{x}$. That is, for all $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathcal{X}$ and $\boldsymbol{y} \in \Omega$,*

$$\left\|\nabla_{\boldsymbol{x}} f(\boldsymbol{y} \mid \boldsymbol{x}_1, \boldsymbol{z}) - \nabla_{\boldsymbol{x}} f(\boldsymbol{y} \mid \boldsymbol{x}_2, \boldsymbol{z})\right\| \leq L_g \|\boldsymbol{x}_1 - \boldsymbol{x}_2\|.$$

This assumption implies that the rate of change of the density function with respect to the decision variable is bounded. It holds for many commonly used distributions. For instance, similar to Assumption 2, this condition is satisfied by Gaussian distributions whose mean and variance are Lipschitz continuous functions of the decision variable.

Assumption 4 (Smooth loss function with bounded gradient)**.** *The loss function $l(\boldsymbol{x}, \boldsymbol{y})$ is Lipschitz continuous and smooth. That is, the following four conditions hold:*

$$\begin{aligned}
&(a) \quad |l(\boldsymbol{x}_1, \boldsymbol{y}) - l(\boldsymbol{x}_2, \boldsymbol{y})| \leq L_1 \|\boldsymbol{x}_1 - \boldsymbol{x}_2\|, \\
&\qquad\qquad \forall \boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathcal{X}, \boldsymbol{y} \in \Omega, \\
&(b) \quad |l(\boldsymbol{x}, \boldsymbol{y}_1) - l(\boldsymbol{x}, \boldsymbol{y}_2)| \leq L_2 \|\boldsymbol{y}_1 - \boldsymbol{y}_2\|, \\
&\qquad\qquad \forall \boldsymbol{x} \in \mathcal{X}, \boldsymbol{y}_1, \boldsymbol{y}_2 \in \Omega, \\
&(c) \quad \|\nabla_{\boldsymbol{x}} l(\boldsymbol{x}_1, \boldsymbol{y}) - \nabla_{\boldsymbol{x}} l(\boldsymbol{x}_2, \boldsymbol{y})\| \leq L_1^c \|\boldsymbol{x}_1 - \boldsymbol{x}_2\|, \\
&\qquad\qquad \forall \boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathcal{X}, \boldsymbol{y} \in \Omega, \\
&(d) \quad \|\nabla_{\boldsymbol{x}} l(\boldsymbol{x}, \boldsymbol{y}_1) - \nabla_{\boldsymbol{x}} l(\boldsymbol{x}, \boldsymbol{y}_2)\| \leq L_2^c \|\boldsymbol{y}_1 - \boldsymbol{y}_2\|, \\
&\qquad\qquad \forall \boldsymbol{x} \in \mathcal{X}, \boldsymbol{y}_1, \boldsymbol{y}_2 \in \Omega,
\end{aligned}$$

where L_1, L_2, L_1^c, L_2^c are corresponding Lipschitz constants.

Assumption 4 articulates the Lipschitz property of the loss function so that we can guarantee the estimation error of the contextual gradient will cause a bounded error in our algorithm. When the value ranges of variables are bounded, many commonly used loss functions, such as linear, quadratic, and hinge loss, naturally satisfy this assumption. From this set of assumptions, it can be seen that both l and $\nabla_{\boldsymbol{x}} l$ have bounded gradients with respect to x and y.

It is worth noting that although we introduce several assumptions to facilitate theoretical analysis, not all of them

are required simultaneously. Each assumption is invoked only where necessary for establishing specific convergence results.

3.3 Statistical properties

The statistical properties of the contextual gradient $\hat{G}(\boldsymbol{x},\boldsymbol{z})$ are deeply rooted in its closeness to the conditional expected gradient of the loss function, that is, $\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},\boldsymbol{z})}[\nabla_{\boldsymbol{x}} l(\boldsymbol{x},\boldsymbol{y})]$. Given a dataset of size N, define

$$\zeta_N = \sup_{\boldsymbol{x},\boldsymbol{z}} \left\| \hat{G}_N(\boldsymbol{x};\boldsymbol{z}) - \mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},\boldsymbol{z})}[\nabla_{\boldsymbol{x}} l(\boldsymbol{x},\boldsymbol{y})] \right\| \tag{12}$$

as the estimation error, which is the maximum gap between the true value of the expected gradient and the contextual gradient for every $\boldsymbol{x}$ and $\boldsymbol{z}$. When the weights $w^{(i)}(\boldsymbol{x},\boldsymbol{z})$ are determined by a proper method, ζ_N diminishes as $N \to \infty$. Lemma 1 establishes that with kNN and kernel methods determining the weights, the contextual gradient (10) is an asymptotically unbiased estimator of the conditional expectation of the gradient of the loss function.

Lemma 1 (Convergence of contextual gradient (Theorem EC.9 in Bertsimas and Kallus, 2019))**.** *Suppose Assumptions 1 and 4 hold, and suppose the joint distribution of* $(\boldsymbol{x},\boldsymbol{z})$ *is absolutely continuous with a density function bounded away from 0 to* ∞ *over the support set and is twice continuously differentiable. Then,*

$$\lim_{N\to\infty} \zeta_N = 0, \tag{13}$$

holds almost surely when

- *the weight functions are determined by the kNN method (5) with* $k = \min\{\lceil C_{kNN} N^{\delta_{kNN}} \rceil, N-1\}$ *for some* $C_{kNN} > 0, 0 < \delta_{kNN} < 1$.
- *the weight functions are determined by the kernel method (6) with* $h = C_{kr} N^{-\delta_{kr}}$ *for some* $C_{kr} > 0, 0 < \delta_{kr} < 1/2d$*, where* $d = \dim(\boldsymbol{x}) + \dim(\boldsymbol{z})$*, denotes the total dimension of the variables influencing the distribution of* $\boldsymbol{y}$.

Beyond the above-mentioned methods, the other two weighting schemes, CART and RF, expressed by equations (8) and (9), fail to achieve convergence under mild assumptions. For CART, $\lim_{N\to\infty} \zeta_N = 0$ holds only under more restrictive assumptions on the training process and data structure. In addition to the previously stated conditions and the bounded tail property, the regression tree must be regular, random-split, and honest; each leaf must contain at least $\min\{\lceil C_{\text{CART}} N^{\delta_{\text{CART}}} \rceil, N-1\}$ training samples for some $C_{\text{CART}} > 0, 0 < \delta_{\text{CART}} < 1$; the historical data $[\boldsymbol{x},\boldsymbol{z}]$ must be uniformly distributed over $[0,1]^d$. As the average result of multiple CARTs, as long as the trees used to construct the RF all meet the above requirements, this RF can also have a similar convergence. Relevant proof is attached in EC.4.1. Although the theoretical conditions for CART and RF are restrictive, our numerical experiments (see Section 5.2) show that these weighting schemes often yield comparable or even superior empirical performance.

The finite-sample guarantee for ζ_N requires additional regularity assumptions on both the expected gradient and the weighting scheme. In EC.1, we provide a finite-sample analysis for the KR method in Example 2. Under the bandwidth choice $h = CN^{-\delta_{kr}}$ with $0 < 2d\delta_{kr} < 1$, the estimation error satisfies

$$O\left(\frac{1}{N^{2\delta_{kr}}} + \sqrt{\frac{\log N}{N^{1-2d\delta_{kr}}}} \right).$$

This bound highlights the joint effect of sample size and problem dimension: the estimation error decreases as N increases, but deteriorates as the dimension d grows. A similar qualitative behavior also holds for other weighting schemes under their corresponding assumptions. We refer readers to Bertsimas and McCord (2018) for detailed finite-sample analyses of CART and RF methods. As for the results of the kNN method, even though there is currently no rigorous theoretical guarantee, numerical experiments as shown in Section 5.2.2 demonstrate that the kNN method also possesses great sample efficiency.

It is important to note that $\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},\boldsymbol{z})}[\nabla_{\boldsymbol{x}} l(\boldsymbol{x},\boldsymbol{y})]$ is not the gradient of the objective (3), which is defined as:

$$G(\boldsymbol{x};\boldsymbol{z}) = \nabla_{\boldsymbol{x}} \mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},\boldsymbol{z})}[l(\boldsymbol{x},\boldsymbol{y})]. \tag{14}$$

When Assumptions 1 and 4 hold, and the joint distribution of $(\boldsymbol{x},\boldsymbol{z})$ is absolutely continuous with a twice continuously differentiable density bounded away from zero and infinity, the wSAA estimator is asymptotically unbiased for the true objective (Bertsimas and Kallus, 2019), that is,

$$\lim_{N\to\infty} \left\| \hat{g}(\boldsymbol{x},\boldsymbol{z}) - \mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},\boldsymbol{z})}[l(\boldsymbol{x},\boldsymbol{y})] \right\| = 0. \tag{15}$$

As such, when the weight functions $w^{(i)}(\boldsymbol{x},\boldsymbol{z})$ are differentiable with respect to $\boldsymbol{x}$, differentiating the left-hand side of (15) with respect to $\boldsymbol{x}$ gives,

$$\begin{aligned} &\lim_{N\to\infty} \left\| \nabla_{\boldsymbol{x}} \hat{g}(\boldsymbol{x},\boldsymbol{z}) - G(\boldsymbol{x};\boldsymbol{z}) \right\| \\ &\quad = \lim_{N\to\infty} \left\| \hat{G}_N(\boldsymbol{x};\boldsymbol{z}) + \sum_{i=1}^{N} \nabla_{\boldsymbol{x}} w^{(i)}(\boldsymbol{x},\boldsymbol{z})\, l(\boldsymbol{x},\boldsymbol{y}^i) - G(\boldsymbol{x};\boldsymbol{z}) \right\| = 0. \end{aligned} \tag{16}$$

Using the reverse triangle inequality, we further derive:

$$\lim_{N\to\infty} \left\| G(\boldsymbol{x};z) - \mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},z)}[\nabla_{\boldsymbol{x}} l(\boldsymbol{x},\boldsymbol{y})] \right\| \tag{17a}$$

$$\geq \lim_{N\to\infty} \left\| G(\boldsymbol{x};z) - \hat{G}_N(\boldsymbol{x};z) \right\| - \lim_{N\to\infty} \left\| \hat{G}_N(\boldsymbol{x};z) - \mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},z)}[\nabla_{\boldsymbol{x}} l(\boldsymbol{x},\boldsymbol{y})] \right\| \tag{17b}$$

$$= \lim_{N\to\infty} \left\| G(\boldsymbol{x};z) - \hat{G}_N(\boldsymbol{x};z) \right\| \tag{17c}$$

$$\geq \lim_{N\to\infty} \left\| \hat{G}_N(\boldsymbol{x};z) + \sum_{i=1}^{N} \nabla_{\boldsymbol{x}} w^{(i)}(\boldsymbol{x},z)\, l(\boldsymbol{x},\boldsymbol{y}^i) - \hat{G}_N(\boldsymbol{x};z) \right\| - \lim_{N\to\infty} \left\| \hat{G}_N(\boldsymbol{x};z) + \sum_{i=1}^{N} \nabla_{\boldsymbol{x}} w^{(i)}(\boldsymbol{x},z)\, l(\boldsymbol{x},\boldsymbol{y}^i) - G(\boldsymbol{x};z) \right\| \tag{17d}$$

$$= \lim_{N\to\infty} \left\| \sum_{i=1}^{N} \nabla_{\boldsymbol{x}} w^{(i)}(\boldsymbol{x},z)\, l(\boldsymbol{x},\boldsymbol{y}^i) \right\|. \tag{17e}$$

The inequality between (17a) and (17b) and that between (17c) and (17d) follow from the reverse triangle inequality. The equality between (17b) and (17c) follows from Lemma 1, while that between (17d) and (17e) follows from equation (16).

These inequalities show that neither the expected gradient $\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},z)}[\nabla_{\boldsymbol{x}} l(\boldsymbol{x},\boldsymbol{y})]$ (corresponding to (17a)) nor the contextual gradient $\hat{G}(\boldsymbol{x};z)$ (corresponding to (17c)) is an unbiased estimator of the gradient of the expected loss function $G(\boldsymbol{x};z)$. From the perspective of the contextual gradient, the discrepancy arises from omitting the term $\sum_{i=1}^{N} \nabla_{\boldsymbol{x}} w^{(i)}(\boldsymbol{x},z)\, l(\boldsymbol{x},\boldsymbol{y}^i)$, which captures part of the decision-dependent structure embedded in the derivative of weighting function. Consequently, an important remaining question is how the contextual gradient (or the expected gradient) relates to the optimality conditions of the original problem, given this deviation.

Theorem 1 bridges the gap between the expected gradient and the optimality of the original problem.

Theorem 1 (Necessary optimality condition)**.** *If Assumptions* 2 *and* 4 *are satisfied, it holds that for any* $\boldsymbol{x} \in \mathcal{X}$*:*

$$\left\| G(\boldsymbol{x},z) - \mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},z)}[\nabla_{\boldsymbol{x}} l(\boldsymbol{x},\boldsymbol{y})] \right\| \leq L_1 \varepsilon.$$

Hence, let $\boldsymbol{x}^*$ *denote an optimal solution to the decision-dependent problem* (3)*. We have*

$$\left\| \mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x}^*,z)}[\nabla_{\boldsymbol{x}} l(\boldsymbol{x}^*,\boldsymbol{y})] \right\| \leq L_1 \varepsilon.$$

Proof Sketch 1. The key idea of the proof is to decompose the difference in expected loss between any decision $\boldsymbol{x}$ and the optimal solution $\boldsymbol{x}^*$ into two components: (i) a *distributional shift* induced by decision-dependent effects, and (ii) a *functional change* due to the variation in the decision itself. Specifically,

$$g(\boldsymbol{x}) - g(\boldsymbol{x}^*) = \underbrace{\left(\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},z)}[l(\boldsymbol{x},\boldsymbol{y})] - \mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x}^*,z)}[l(\boldsymbol{x},\boldsymbol{y})]\right)}_{\text{Distributional shift}} + \underbrace{\left(\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x}^*,z)}[l(\boldsymbol{x},\boldsymbol{y})] - \mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x}^*,z)}[l(\boldsymbol{x}^*,\boldsymbol{y})]\right)}_{\text{Functional change}}.$$

The distributional-shift term is bounded using the Kantorovich–Rubinstein duality, which links it to the Wasserstein-1 distance and yields a bound proportional to the sensitivity parameter ε under Assumption 2. The functional-change term is controlled via a first-order Taylor expansion, which relates it to the expected gradient of the loss function. Combining these bounds and taking limits leads to the desired result in Theorem 1. □

Theorem 1 shows that the expected gradient $\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},z)}[\nabla_{\boldsymbol{x}} l(\boldsymbol{x},\boldsymbol{y})]$ has a bounded gap from the true gradient of the objective function and must be bounded in norm at the optimal solution. The bound is proportional to the sensitivity parameter ε. This result implies that the expected gradient is a good metric of optimality when the sensitivity is low. Moreover, since the contextual gradient $\hat{G}(\boldsymbol{x};z)$ serves as an asymptotically unbiased estimator of the expected gradient (Lemma 1), it is subject to the same bound as $N \to \infty$. This observation offers two important insights.

First, it quantifies the portion of gradient information that is neglected by the contextual gradient at the optimal solution. According to the inequalities in (17), we further obtain

$$\lim_{N\to\infty} \left\| \sum_{i=1}^{N} \nabla_{\boldsymbol{x}} w^{(i)}(\boldsymbol{x},z)\, l(\boldsymbol{x},\boldsymbol{y}) \right\| \leq L_1 \varepsilon, \tag{18}$$

when the weights are differentiable. This result implies that the loss of gradient information at the optimal solution is uniformly bounded and thus limited in magnitude.

Second, it establishes the contextual gradient as a reliable optimality indicator that effectively distinguishes near-optimal solutions. In other words, a small contextual-gradient norm signifies proximity to the true optimum. This property enables the contextual gradient to be incorporated into standard gradient-based optimization frameworks—including GD, SGD, and momentum-based variants—as a computationally efficient proxy for the true gradient (14). Example 8 illustrates this property through a simple example.

Example 8 (Contextual gradient as a screening condition)**.** *Consider a toy example with loss function* $l(x,y) = -(x_1 y_1 + x_2 y_2)$*, where* $y_1 \sim U(a_1 - b_1 x_1 + z, a_1 - b_1 x_1 + z + 1)$*,* $y_2 \sim U(a_2 - b_2 x_2 + z, a_2 - b_2 x_2 + z + 1)$*, and* $x_1, x_2 \geq 0$*. In this setting,* $\varepsilon = \|(b_1, b_2)\|$*,* $L_1 = \|(a_1 + z + 1, a_2 + z + 1)\|$*,*

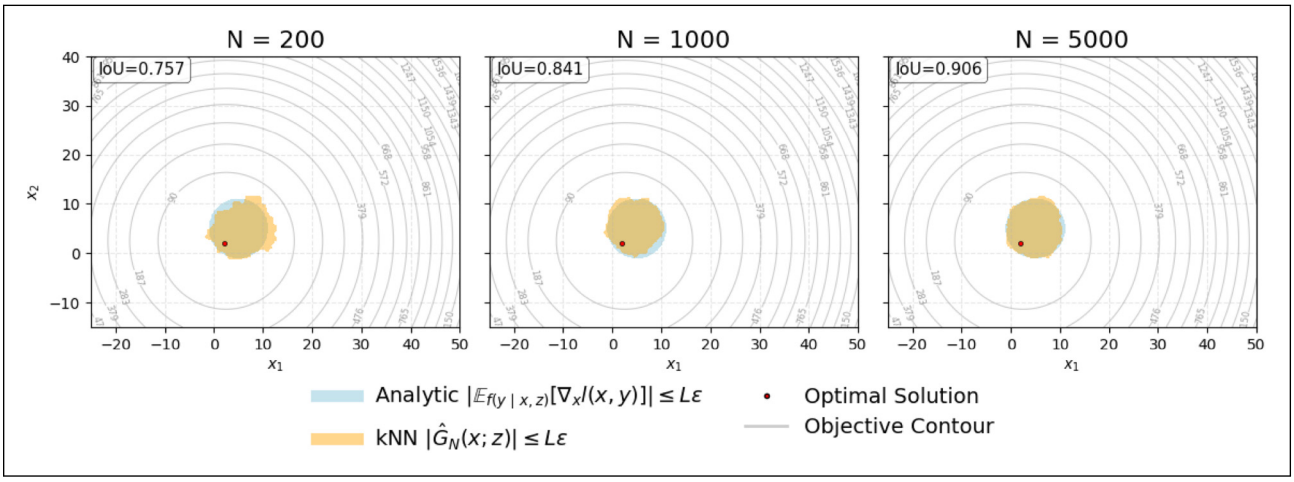


Figure 1. Analytic versus k-nearest neighbor (kNN) estimated screening regions for different sample sizes.

and the optimal solution is $\boldsymbol{x}^* = \arg\min_{\boldsymbol{x}} \mathbb{E}[l(\boldsymbol{x}, y)] = \big((a_1 + z)/2b_1, (a_2 + z)/2b_2\big)$.

Figure 1 shows the screening property of the contextual gradient for $a_1 = a_2 = 2$, $b_1 = b_2 = 0.5$, $z = 0$, *and* $N \in \{200, 1000, 5000\}$. *The gray contours indicate objective values, and the red dot marks the optimal solution. The blue region corresponds to points satisfying* $\|\mathbb{E}_{f(y|x,z)}[\nabla_x l(x, y)]\| \le L_1\varepsilon = 3$, *while the yellow region satisfies* $\|\hat{G}_N(x; z)\| \le L_1\varepsilon$ *using kNN weights* (5) *with* $k = \lceil 0.4N^{0.5} \rceil$. *The intersection-over-union between the two regions is reported in each panel. As shown, even with limited samples, both regions concentrate around high-quality solutions and contain the optimum. This demonstrates that the contextual gradient provides an effective screening or stopping criterion for CSO problems with decision-dependent effects.*

4 CGD algorithm and its convergence analysis

4.1 CGD algorithm

In this section, we incorporate the contextual gradient into a GD framework and propose the CGD algorithm shown in Algorithm 1. In this section, $\Pi_{\mathcal{X}}(\cdot) : \mathbb{R}^d \to \mathcal{X}$ denotes the Euclidean projection onto the feasible set $\mathcal{X}$. The CGD algorithm follows a similar iterative scheme to classical GD. The stopping criterion could be when $|\hat{G}_N(\boldsymbol{x}^r; z)|$ is smaller than a certain value or when r reaches the maximum iterations. Details of the step-size strategies used in our implementation are discussed in EC.3.

4.2 Convergence analysis

Since the CGD algorithm relies on the contextual gradient rather than the true gradient of the expected loss, a natural question is the quality of the resulting solution. For gradient-based methods, convergence to the global optimum generally requires convexity. We therefore begin our analysis by assuming that the loss function $l(\boldsymbol{x}, \boldsymbol{y})$ is convex in $\boldsymbol{x}$. This assumption is relatively mild, as it is imposed only on the sample-wise loss function rather than on the expected objective, which involves integration over the joint distribution of the uncertainty parameters and contextual covariates. In this section, we denote the solution obtained at the r-th iteration of Algorithm 1 by $\boldsymbol{x}_N^r$ and let η_r denote the step size at the r-th iteration of Algorithm 1.

Theorem 2 quantifies the optimality gap between the CGD iterations and the true optimal solution.

Theorem 2 (Error bound in the convex case)**.** *Suppose* $l(\boldsymbol{x}, \boldsymbol{y})$ *is convex in* $\boldsymbol{x}$, *and Assumptions* 1, 2, 4*(a), and* 4 *(b) hold. Then, after k iterations:*

$$\min_{0\le r\le k}\left\{\mathbb{E}_{f(y|\boldsymbol{x}_N^r,z)}[l(\boldsymbol{x}_N^r,\boldsymbol{y})] - \mathbb{E}_{f(y|\boldsymbol{x}^*,z)}[l(\boldsymbol{x}^*,\boldsymbol{y})]\right\} \le \frac{\zeta_N^2\sum_{r=0}^k(\eta_r)^2 + 2\zeta_N\sum_{r=0}^k\eta_r\|\boldsymbol{x}_N^r - \boldsymbol{x}^*\|}{2\sum_{r=0}^k\eta_r} + \frac{\|\boldsymbol{x}_N^0 - \boldsymbol{x}^*\|^2 + L_1^2\sum_{r=0}^k(\eta_r)^2}{2\sum_{r=0}^k\eta_r} + \frac{\varepsilon L_2\sum_{r=0}^k\eta_r\|\boldsymbol{x}_N^r - \boldsymbol{x}^*\|}{\sum_{r=0}^k\eta_r}. \tag{19}$$

Algorithm 1. Contextual gradient descent (CGD) algorithm

Input: Initial solution $\boldsymbol{x}^0$, covariate $\boldsymbol{z}$, dataset $\{(\boldsymbol{x}^i, \boldsymbol{z}^i, \boldsymbol{y}^i)\}_{i=1}^N$
Output: Final solution $\hat{\boldsymbol{x}}^*$

```
Set iteration counter r = 0
while Stopping criterion is not met do
    Compute contextual gradient Ĝ_N(x^r; z) using (10)
    Select step size η_r
    Update solution: x^{r+1} = Π_X(x^r − η_r Ĝ_N(x^r; z))
    Increment iteration: r ← r + 1
end while
return x̂* = x^r
```

Here, L_1 and L_2 are the upper bound on $\|\nabla_{\boldsymbol{x}} l(\boldsymbol{x},\boldsymbol{y})\|$ and $\|\nabla_{\boldsymbol{y}} l(\boldsymbol{x},\boldsymbol{y})\|$, respectively, and ε is the sensitivity parameter from Assumption 2. ζ_N is the estimation error defined in (12).

Proof Sketch 2. The proof of Theorem 2 first leverages the relationship between the contextual gradient and the expected gradient to connect the convergence of the CGD algorithm with the expected gradient. It then uses the convexity of $l(\boldsymbol{x},\boldsymbol{y})$ to relate the optimality gap to the accumulated expected gradients over iterations, and applies the Kantorovich–Rubinstein inequality to bound the impact of decision dependence through the sensitivity parameter ε. □

The three terms on the right-hand side of (19) correspond to distinct sources of error:

- **Estimation error:** The first term captures the additional gap introduced by the estimation error of the contextual gradient in approximating the expected gradient of the loss function. A smaller estimation error ζ_N leads to a smaller value of this term. For the weighting schemes described in Lemma 1, this error term diminishes as the sample size $N \to \infty$. When the dataset is finite, the first component of the numerator still vanishes asymptotically, provided that the step-size sequence satisfies the classical diminishing conditions

$$\sum_{r=0}^{\infty} \eta_r = \infty, \quad \frac{\sum_{r=0}^{k} (\eta_r)^2}{\sum_{r=0}^{k} \eta_r} \to 0 \quad \text{as } k \to \infty.$$

  A typical diminishing step-size policy satisfying these conditions is

$$\eta_r = \frac{\bar{\eta}}{(r + k_0)^{\alpha}},$$

  where $\bar{\eta} > 0$ is the initial learning rate, $0.5 < \alpha \leq 1$, and $k_0 > 0$ is an offset parameter preventing excessively large initial steps. In contrast, the second component diminishes only when the iterate sequence $\{\boldsymbol{x}_N^r\}_{r=0}^{\infty}$ converges sufficiently close to the optimal solution.
- **Optimization error:** The second term is the standard optimization error associated with GD methods. Under a constant step size η, this term converges to $L_1^2\eta/2$ as $k \to \infty$, and can therefore be made arbitrarily small by choosing a sufficiently small step size. It vanishes entirely when the step-size sequence satisfies the diminishing conditions above.
- **Decision-dependent error:** The third term arises from the decision-dependent effect. When $\varepsilon = 0$, this term vanishes, reducing the analysis to the standard convergence result for GD applied directly to the wSAA objective. When $\varepsilon > 0$, however, convergence additionally depends on whether the iterates $\boldsymbol{x}_N^r$ approach the optimal solution $\boldsymbol{x}^*$ sufficiently closely.

In conclusion, although some components of the bound vanish under standard diminishing step-size conditions, two terms additionally depend on the convergence behavior of the iterate sequence itself. Specifically, the scales of both the second component of the estimation error term and the decision-dependent error term are determined by whether the solution sequence $\{\boldsymbol{x}_N^r\}_k$ converges to a small neighborhood of the optimal solution $\boldsymbol{x}^*$.

However, such convergence cannot generally be guaranteed in the convex setting. An extreme example is the pop-up store placement problem in Example 6, where the loss function does not explicitly depend on the decision variable $\boldsymbol{x}$. In this case, the contextual gradient is identically zero for all $\boldsymbol{x}$, and the CGD algorithm degenerates into a stationary process that remains at its initial point. As such, it is important to identify conditions under which the solution sequence converges to a neighborhood of the optimal solution. In the remainder of this section, we show that sufficiently strong convexity of the loss function provides such a condition by ensuring that the contextual gradient retains enough information about the true descent direction.

4.2.1 Convergence under the strongly convex case. In this subsection, we focus on the loss function satisfying the condition that for all $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathcal{X}$, we have

$$l(\boldsymbol{x}_2,\boldsymbol{y}) \geq l(\boldsymbol{x}_1,\boldsymbol{y}) + \nabla_{\boldsymbol{x}} l(\boldsymbol{x}_1,\boldsymbol{y})^{\top}(\boldsymbol{x}_2 - \boldsymbol{x}_1) + \frac{\gamma}{2}\|\boldsymbol{x}_2 - \boldsymbol{x}_1\|^2,$$

where γ is a coefficient measuring the strength of strong convexity. Under the strongly convex condition, we establish convergence through an intermediate concept called the *stable point* as Mendler-Dünner et al. (2020), formally defined as follows.

Definition 2 (Stable point)**.** *Given a covariate $\boldsymbol{z}$, a point $\boldsymbol{x}_{PS}$ is called a stable point if it satisfies the following fixed-point condition:*

$$\boldsymbol{x}_{PS} = \arg\min_{\boldsymbol{x}\in\mathcal{X}} \mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x}_{PS},\boldsymbol{z})}[l(\boldsymbol{x},\boldsymbol{y})]. \tag{20}$$

Intuitively, a stable point represents a decision that minimizes the expected loss under its own induced conditional distribution of outcomes. Theorem 3 confirms the convergence of the CGD algorithm to the optimal solution through the stable point.

Theorem 3 (Convergence under the strongly convex case with constant step size)**.** *Suppose Assumptions* 1, 2, *and* 4 *hold. Define*

$$A = \gamma - \varepsilon L_1^c,\ B = L_1^c\sqrt{1+\varepsilon^2},\ C = \sqrt{1 - \eta A + \eta^2 B^2} < 1,$$

and

$$UB(\eta,\zeta_N)=\frac{\zeta_N}{2(A-\eta B^2)}\times\left(1+\sqrt{2}\eta B+\sqrt{1+2A\eta+2\sqrt{2}\eta B}\right).$$

When $A>0$, there exists a unique stable point $\boldsymbol{x}_{PS}$ and if a constant step size η satisfying $0<2B^2\eta^2-2A\eta+1<1$ is used, then the following properties hold:

- *if $\|\boldsymbol{x}_N^k-\boldsymbol{x}_{PS}\|>UB(\eta,\zeta_N)$, then*

$$\|\boldsymbol{x}_N^k-\boldsymbol{x}_{PS}\|\geq\|\boldsymbol{x}_N^{k+1}-\boldsymbol{x}_{PS}\|;$$

- *if $\|\boldsymbol{x}_N^k-\boldsymbol{x}_{PS}\|\leq UB(\eta,\zeta_N)$, then for all $k'>k$,*

$$\|\boldsymbol{x}_N^{k'}-\boldsymbol{x}_{PS}\|\leq UB(\eta,\zeta_N).$$

Moreover, the distance between the iterate and the stable points satisfies

$$\|\boldsymbol{x}_N^k-\boldsymbol{x}_{PS}\|\leq\max\left\{C^k\|\boldsymbol{x}_N^0-\boldsymbol{x}_{PS}\|,\ 2UB(\eta,\zeta_N)\right\}\tag{21}$$

and that between the iterate and the optimal solution satisfies

$$\|\boldsymbol{x}_N^k-\boldsymbol{x}^*\|\leq\max\left\{C^k\|\boldsymbol{x}_N^0-\boldsymbol{x}_{PS}\|,\ 2UB(\eta,\zeta_N)\right\}+\frac{2L_2\varepsilon}{\gamma}.\tag{22}$$

Proof Sketch 3. The proof builds on the relationship between the contextual gradient and the expected gradient. Under similar assumptions, Perdomo et al. (2020) establish linear convergence of GD using the expected gradient to a stable point. By linking the contextual gradient to the expected gradient, we extend their results to contextual settings and additionally quantify the estimation error induced by the contextual gradient. □

Theorem 3 shows that when $A>0$ and $\|\boldsymbol{x}_N^k-\boldsymbol{x}_{PS}\|\geq 2UB(\eta,\zeta_N)$, the upper bound on the distance between the current and the stable point as well as the optimal solution decreases exponentially over iterations. This result characterizes the conditions under which the contextual gradient remains sufficiently aligned with the true descent direction, thereby enabling the CGD algorithm to efficiently converge to a neighborhood of the optimal solution. These two conditions characterize when the contextual gradient can withstand different sources of bias from the true descent direction.

- The condition $A>0$ is an intrinsic property of the problem, requiring that the strong convexity of the loss function be sufficiently large relative to the magnitude of the decision-dependent effect. Intuitively, this condition ensures that the retained component of the contextual gradient dominates the neglected one. Specifically, by strong convexity, for any $\boldsymbol{x}_1,\boldsymbol{x}_2$ and any distribution $f(\boldsymbol{y}\mid\boldsymbol{x}',z)$,

$$\begin{aligned}&\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x}',z)}[l(\boldsymbol{x}_2,\boldsymbol{y})]\\&\quad\geq\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x}',z)}[l(\boldsymbol{x}_1,\boldsymbol{y})]+\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x}',z)}[\nabla_{\boldsymbol{x}}l(\boldsymbol{x}_1,\boldsymbol{y})]^\top(\boldsymbol{x}_2-\boldsymbol{x}_1)\\&\qquad+\frac{\gamma}{2}\|\boldsymbol{x}_2-\boldsymbol{x}_1\|^2.\end{aligned}$$

Since $f(\boldsymbol{y}\mid\boldsymbol{x}',z)$ is independent of $\boldsymbol{x}$, differentiation and expectation can be interchanged, implying that $\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x}',z)}[l(\boldsymbol{x},\boldsymbol{y})]$ is strongly convex in $\boldsymbol{x}$ for any fixed $\boldsymbol{x}'$. Defining

$$H(\boldsymbol{x}):=\arg\min_{\boldsymbol{x}'\in\mathcal{X}}\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},z)}[l(\boldsymbol{x}',\boldsymbol{y})],$$

strong convexity yields

$$\frac{\gamma}{2}\|\boldsymbol{x}-H(\boldsymbol{x})\|\leq\left\|\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},z)}[\nabla_{\boldsymbol{x}}l(\boldsymbol{x},\boldsymbol{y})]\right\|.\tag{23}$$

Therefore, whenever $\boldsymbol{x}$ is not a stable point, that is, $\boldsymbol{x}\neq H(\boldsymbol{x})$, the magnitude of the expected gradient grows proportionally with the strong convexity parameter γ. Lemma 1 shows that the neglected component of the wSAA gradient is uniformly bounded by $L_1\varepsilon$. Hence,

$$\begin{aligned}\frac{\frac{\gamma}{2}\|\boldsymbol{x}-H(\boldsymbol{x})\|}{L_1\varepsilon}&\leq\frac{\left\|\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},z)}[\nabla_{\boldsymbol{x}}l(\boldsymbol{x},\boldsymbol{y})]\right\|}{\left\|G(\boldsymbol{x},z)-\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x},z)}[\nabla_{\boldsymbol{x}}l(\boldsymbol{x},\boldsymbol{y})]\right\|}\\&=\lim_{N\to\infty}\frac{\|\hat{G}_N(\boldsymbol{x};z)\|}{\left\|\sum_{i=1}^N\nabla_{\boldsymbol{x}}w^{(i)}(\boldsymbol{x},z)\,l(\boldsymbol{x},\boldsymbol{y}^i)\right\|}.\end{aligned}\tag{24}$$

This relation shows that stronger convexity amplifies the informative component of the contextual gradient relative to the neglected decision-dependent component, thereby ensuring alignment with the true descent direction.

- The condition $\|\boldsymbol{x}_N^k-\boldsymbol{x}_{PS}\|\geq 2UB(\eta,\zeta_N)$ reflects the influence of the estimation quality of the contextual gradient on convergence. As shown in the proof of Theorem 3 in EC.4.4, the magnitude of the expected gradient is proportional to the distance between the current iterate and the stable point:

$$\|\mathbb{E}_{f(\boldsymbol{y}|\boldsymbol{x}_N^k,z)}[\nabla_{\boldsymbol{x}}l(\boldsymbol{x}_N^k,\boldsymbol{y})]\|\geq A\|\boldsymbol{x}_N^k-\boldsymbol{x}_{PS}\|.$$

As such, for a given level of estimation error ζ_N, when the iterate is sufficiently far from the stable point, the signal contained in the expected gradient dominates the estimation noise. Consequently, the contextual gradient

remains sufficiently aligned with the true descent direction, allowing the CGD algorithm to continue making effective progress toward the stable point.

Moreover, when $k \to \infty$, the right-hand side of inequality (22), $2UB(\eta, \zeta_N) + (2L_2\varepsilon)/\gamma$, reflects the influence of the total bias between the contextual gradient and the true gradient on the convergence behavior of the CGD algorithm. The first component captures the estimation error introduced by the weighting scheme. Since $2UB(\eta, \zeta_N)$ is proportionate to ζ_N, under certain conditions, it shrinks at the same rate as ζ_N and vanishes as the sample size goes to infinity, as discussed in Section 3.3.

The second component arises from the decision-dependent effect and quantifies the information loss caused by neglecting part of the gradient of the wSAA objective. Importantly, the explicit form of this residual shows that the resulting gap decreases as the loss function becomes more strongly convex. This result indicates that strong convexity can partially compensate for the information omitted by the contextual gradient, thereby ensuring reliable optimization performance despite the biased descent direction. This compensatory effect can be understood by comparing the relative magnitudes of the contextual gradient and the neglected component of the wSAA gradient shown in equation (24).

Remark 1 (Optimality of the convergence rate of CGD algorithm)**.** A*s k increases, the solution of the CGD algorithm converges to a neighborhood of the optimal solution with radius* $2UB(\eta, \zeta_N) + 2L_2\varepsilon/\gamma$ *at a linear rate C with a constant step* η *satisfying* $0 < 2B^2\eta^2 - 2A\eta + 1 < 1$. *It indicates that the resulting rate is optimal. By the classical complexity results of Nesterov (2013), the best attainable rate for first-order methods without decision-dependent effect is linear. Since introducing decision dependence only increases the difficulty of the problem, its convergence rate cannot surpass the linear benchmark. Our results attain this benchmark in the presence of decision dependence, demonstrating that the CGD algorithm achieves the optimal convergence rate. It is worth noting that linear convergence cannot be guaranteed when applying GD directly to the wSAA objective, even if that objective is differentiable. This is because the strong convexity of the loss function does not imply strong convexity of the corresponding wSAA objective. In EC.2, we provide an example of such a case.*

With the desirable convergence behavior of the solution, we can guarantee the optimality gap between the objectives of the obtained solution and the optimal one.

Corollary 1 (Error bound under strong convexity with constant step size)**.** *Suppose Assumptions* 1*,* 2*, and* 4 *hold and* $A > 0$. *When constant step size* η *satisfying* $0 < 2B^2\eta^2 - 2A\eta + 1 < 1$ *is used, we have*

$$\begin{aligned}
&\min_{0\le r\le k}\left\{\mathbb{E}_{f(y|x_N^r,z)}[l(x_N^r,y)] - \mathbb{E}_{f(y|x^*,z)}[l(x^*,y)]\right\} \\
&\quad\le \frac{\|x_N^0 - x^*\|^2 + (L_1^2+\zeta_N^2)(k+1)(\eta)^2}{2(k+1)\eta} \\
&\qquad + \frac{(\varepsilon L_2+\zeta_N)\sum_{r=0}^{k} \times\left(\max\left\{C^k\|x_N^0 - x_{PS}\|,\ 2UB(\eta,\zeta_N)\right\} + \frac{2L_2\varepsilon}{\gamma}\right)}{(k+1)}.
\end{aligned}$$

With $k \to \infty$, *we have*

$$\begin{aligned}
&\lim_{k\to\infty}\min_{0\le r\le k}\left\{\mathbb{E}_{f(y|x_N^r,z)}[l(x_N^r,y)] - \mathbb{E}_{f(y|x^*,z)}[l(x^*,y)]\right\} \\
&\quad\le (L_1^2+\zeta_N^2)\eta + 2(\varepsilon L_2+\zeta_N)\left(UB(\eta,\zeta_N) + \frac{L_2\varepsilon}{\gamma}\right).
\end{aligned}$$

Moreover, for the weighting methods satisfying the conditions in Lemma 1, as $N \to \infty$ ($\zeta_N \to 0$ *almost surely), we obtain*

$$\begin{aligned}
&\lim_{N\to\infty}\lim_{k\to\infty}\min_{0\le r\le k}\left\{\mathbb{E}_{f(y|x_N^r,z)}[l(x_N^r,y)] - \mathbb{E}_{f(y|x^*,z)}[l(x^*,y)]\right\} \\
&\quad\le L_1^2\eta + 2\varepsilon^2L_2^2/\gamma.
\end{aligned}$$

As discussed earlier, when a constant step size is used, the final error bound of the algorithm includes a term that depends on the step size η. This term becomes negligible when η is chosen to be sufficiently small. Moreover, this term can be completely eliminated by adopting a diminishing step-size sequence that satisfies $\sum_{r=0}^{\infty}\eta_r = \infty$ and $\sum_{r=0}^{k}\eta_r^2/\sum_{r=0}^{k}\eta_r \to 0$ as $k \to \infty$. Theorem 4 establishes the asymptotic convergence behavior of the CGD algorithm under such diminishing step sizes.

Theorem 4 (Asymptotic convergence under strong convexity with diminishing step size)**.** *Suppose Assumptions 1, 2, and 4 hold, and* $A > 0$. *If a diminishing step-size sequence* $\{\eta_k\}_{k=0}^{\infty}$ *satisfying* $\eta_k > \eta_{k+1}, \lim_{k\to\infty}\eta_k = 0, \sum_{k=0}^{\infty}\eta_k = \infty$ *is used, and the initial step size satisfies* $0 < 2B^2\eta_0^2 - 2A\eta_0 + 1 < 1$, *then*

$$\lim_{K\to\infty}\sup_{k\ge K}\|x_N^k - x^*\| \le \frac{\zeta_N}{A} + \frac{2L_2\varepsilon}{\gamma}, \tag{25}$$

As $k \to \infty$,

$$\begin{aligned}
&\lim_{k\to\infty}\min_{0\le r\le k}\left\{\mathbb{E}_{f(y|x_N^r,z)}[l(x_N^r,y)] - \mathbb{E}_{f(y|x^*,z)}[l(x^*,y)]\right\} \\
&\quad\le 2(\varepsilon L_2+\zeta_N)\left(\frac{\zeta_N}{A} + \frac{L_2\varepsilon}{\gamma}\right).
\end{aligned} \tag{26}$$

Moreover, for weighting methods that satisfy the conditions in Lemma 1, as $N \to \infty$, *we obtain*

$$\lim_{N\to\infty}\lim_{k\to\infty}\min_{0\le r\le k}\left\{\mathbb{E}_y[l(x_N^r,y)] - \mathbb{E}_y[l(x^*,y)]\right\} \le 2\varepsilon^2L_2^2/\gamma.$$

Proof Sketch 4. The proof of this theorem is consistent with the general ideas of Theorems 2 and 3. In this process, the summation of the series of step sizes needs to be handled. In this paper, the integral method is adopted to scale and process it. □

This result shows that adopting a diminishing step-size rule yields a tighter asymptotic convergence bound. However, diminishing step sizes do not provide a nonasymptotic guarantee and therefore do not ensure a convergence bound within a finite number of iterations.

Although the examples in Section 1.2 fail to satisfy strong convexity in their original form, their strongly convex counterpart naturally arise from practical requirements. In the following, we illustrate how common practical requirements on the problems in Section 1.2 cause the strongly convex extensions of the examples.

Example 9 (Price-setting newsvendor problem with price stability)**.** *In practice, maintaining stable prices and order quantities is essential for operational consistency. To enforce this stability requirement, quadratic regularization terms are introduced into the loss function. Specifically, we have*

$$l_{pen}(\boldsymbol{x}, y) = -(p-c)q+(p-s)(q-d)^{+}+\lambda_p(p-p_{sd})^2+\lambda_q(q-q_{sd})^2,$$

where λ_p and λ_q are regularization coefficients, and p_{sd}, q_{sd} are reference values.

Example 10 (Pop-up store location considering mobile cost)**.** *It is critical to account for travel costs in mobile retailing, as transportation incurs additional expenses such as fuel consumption, labor time, and logistics overhead. Suppose that stores typically operate from a central depot denoted as $\boldsymbol{x}_0 \in \mathbb{R}^2$, and define the travel cost as $c_{travel}(\boldsymbol{x}) = \lambda \cdot \|\boldsymbol{x}-\boldsymbol{x}_0\|^2$, where $\lambda > 0$ is a cost coefficient and the squared Euclidean distance captures increasing marginal travel cost. The resulting loss function is*

$$l(\boldsymbol{x}, \boldsymbol{y}) = -p \cdot y + \lambda \cdot \|\boldsymbol{x} - \boldsymbol{x}_0\|^2.$$

Example 11 (Portfolio management with diversity requirement)**.** *In portfolio management, diversification is a fundamental requirement for risk control. Portfolios that concentrate heavily on a small number of assets are highly vulnerable to idiosyncratic shocks, leading to unstable and potentially severe losses. To mitigate this risk, managers usually deliberately diversify their portfolios by introducing an ℓ_2-norm regularization term. The resulting loss function becomes*

$$l(\boldsymbol{x}, \boldsymbol{y}, \eta) = -\boldsymbol{x}^{\top}\boldsymbol{y} + \rho \cdot \left(\eta + \frac{1}{1-\alpha}\left(-\boldsymbol{x}^{\top}\boldsymbol{y} - \eta\right)^{+}\right) + \lambda \cdot \|\boldsymbol{x}\|^2,$$

where $\lambda > 0$ is the diversification parameter.

Although convergence to the global optimum can no longer be guaranteed in general convex and nonconvex problems, the algorithm still exhibits desirable stationary-point convergence properties under suitable conditions. This is also the strongest guarantee one can generally expect from ordinary GD methods in nonconvex optimization. We refer interested readers to EC.5 for both the theoretical and numerical analysis of the convergence of the CGD algorithm to stationary points of $\mathbb{E}_{f(y|x,z)}[\nabla_{\boldsymbol{x}} l(\boldsymbol{x}, \boldsymbol{y})]$ in the general convex and nonconvex settings.

5 Numerical results

In this section, we evaluate the convergence behavior, solution quality, and computational efficiency of the proposed CGD algorithm using synthetic data. We first describe the experimental setup in Section 5.1. Next, in Section 5.2, we investigate the convergence behavior of CGD under different weighting schemes and parameter settings. Finally, in Section 5.3, we compare the efficiency and effectiveness of CGD with several alternative approaches.

5.1 Experiment setup

We demonstrate the effectiveness of the proposed CGD algorithm using the price-setting newsvendor problem with stabilizing regularization terms introduced in Example 9. The resulting loss function is strongly convex, which aligns with the theoretical setting studied in this paper. In the experiments, we set the reference points $p_{\text{sd}} = 60$ and $q_{\text{sd}} = 100$ and the salvage value s is set as 0. The value space is set as $p \in [10, 200], q \in [0, 200]$, satisfying the consistent bound requirement in Assumption 1. To satisfy the smoothness requirement in Assumption 4, we adopt a Huber-type smoothing technique that has been widely used in the literature (Chen and Mangasarian, 1996). Specifically, we replace $(x)_+$ with

$$S_\delta(x) = \begin{cases} 0, & x \le 0, \\ \dfrac{x^2}{2\delta}, & 0 < x < \delta, \\ x - \dfrac{\delta}{2}, & x \ge \delta, \end{cases}$$

and approximate the original loss function by

$$l^{\delta}_{\text{pen}}(\boldsymbol{x}; y) = -(p-c)q + (p-s)S_\delta(q-d) + \lambda_p(p-p_{\text{sd}})^2 + \lambda_q(q-q_{\text{sd}})^2.$$

This smoothing introduces a controlled approximation error. In particular, for all $x \in \mathbb{R}$, $0 \le (x)_+ - S_\delta(x) \le \delta/2$. Consequently, the approximation error of the smoothed objective is uniformly bounded by $(p-s)\delta/2$. Therefore, by choosing δ sufficiently small, the smoothed formulation can

approximate the original problem arbitrarily well while preserving differentiability and numerical stability. In addition, its quadratic nature ensures the satisfaction of Assumption 4.

Our data generation process closely follows the setup in Lin et al. (2022). The contextual features z are independently drawn from a four-dimensional Gaussian distribution $N(0, \Sigma)$, where Σ is a diagonal matrix with entries $1, 2, 3, 4$. Prices are independently sampled from a uniform distribution $U(10, 200)$. For the demand, we consider two types of demand models:

- **Homoscedastic model:** The first model assumes that the decision (price) affects only the mean of demand, which is used in EC.1 of Lin et al. (2022). Specifically, demand is generated as

$$D = \max\{0, 60 - \varepsilon p + 12a^T(z + 0.25\phi) + 5b^T z\theta\}, \quad (27)$$

  where $\phi \sim N(0, I)$ is a four-dimensional random vector, I is the identity matrix, $\theta \sim N(0, 1)$ is a scalar random variable, $a = (0.8, 1, 1, 1)^T$, and $b = (-1, 1, 0, 0)^T$.
- **Heteroscedastic model:** The second model introduces heteroscedasticity by allowing the variance of demand to depend on the price, which is similar to the setting in Harsha et al. (2021)

$$D = \max\{0, 60 - \varepsilon_1 p + 12a^T z + \varepsilon_2^2 p a^T \phi + 5b^T z\theta\}, \quad (28)$$

  where ε_1 and ε_2 are constants. The introduction of price-dependent variance violates many common structural assumptions in previous literature and increases the complexity of the learning task.

Note that both cases satisfy Assumption 2. The homoscedastic model satisfies the ε-sensitivity condition directly, while the heteroscedastic model is $\max\{|\varepsilon_1|, |\varepsilon_2|\}$-sensitive. Moreover, since D follows a distribution whose mean and variance are Lipschitz continuous functions of the decision variable, both settings also satisfy Assumption 3. Therefore, all assumptions required for the theoretical analysis are satisfied.

The weights are determined using ML methods listed in Examples 1–4. The hyperparameters associated with each weighting scheme are tuned via grid search. The initial solution of the CGD algorithm is randomly selected from the feasible region. The algorithm terminates when the ℓ_2-norm of the contextual gradient falls below 0.01 or when the maximum number of iterations reaches 150.

We evaluate the performance of each method using the *optimality gap*, defined as:

$$\text{optimality gap} = \frac{\left|\mathbb{E}_{f(y|x^*,z)}[l(x^*, y)] - \mathbb{E}_{f(y|x,z)}[l(x, y)]\right|}{\left|\mathbb{E}_{f(y|x^*,z)}[l(x^*, y)]\right|}.$$

Here, x^* denotes the optimal solution under the true data-generating distribution, which is obtained by enumeration in the experiments. All experiments are conducted on a FusionServer G5500 V6 platform with 128 CPU cores and 2 TB RAM, running Ubuntu 22.04.4 LTS on Linux x86_64.

5.2 Convergence performance

In this section, we evaluate the convergence behavior of the CGD algorithm from multiple perspectives. Section 5.2.1 investigates the convergence speed of CGD and the post-convergence optimality gap under different values of the critical ratio $L_2\varepsilon/\gamma$. Section 5.2.2 examines the impact of sample size on solution quality.

5.2.1 Convergence under different critical ratios. We first examine the convergence behavior of the CGD algorithm under different weighting strategies for the price-setting newsvendor problem. In this section, the sample size is fixed at $N = 1000$. Since the homoscedastic and heteroscedastic demand models exhibit qualitatively similar behaviors, we only report the results for the heteroscedastic setting for brevity. Accordingly, the sensitivity parameter ε is defined in equation (28). In the regularized price-setting newsvendor problem, the strong convexity parameter is given by $\gamma = \min\{\lambda_p, \lambda_q\}$, while L_2 is bounded above by $\max\{p, s\} = 200$ under our experimental setting. In the following experiments, we fix $\lambda_p = \lambda_q = 1$ (and thus $\gamma = 1$), and vary ε from 0.01 to 0.4. Consequently, the ratio $L_2\varepsilon/\gamma$ ranges from 2 to 80. This setup allows us to empirically investigate how the relative magnitude of the decision-dependent effect and the strong convexity influences the convergence behavior and solution quality of the CGD algorithm.

Figure 2 illustrates the convergence behavior of the CGD algorithm for a randomly generated context vector z. The figure reports the optimality gaps over iterations under different values of the critical ratio and various weighting strategies. As shown, the optimality gap initially decreases at an exponential rate before stabilizing around a neighborhood of the optimal solution. Moreover, the size of the residual gap after convergence is strongly influenced by the critical ratio. To further examine this effect, Figure 3 presents the average final optimality gaps under different critical ratios. As the figure shows, the final gaps consistently increase with the critical ratio, empirically validating the theoretical relationship between the critical ratio and the convergence accuracy of the CGD algorithm.

Overall, these results are consistent with the theoretical analysis in Theorem 3. In particular, when the sample size is sufficiently large (e.g., 1,000 samples in this experiment), the CGD algorithm rapidly converges to solutions with stable and bounded optimality gaps under different weighting schemes, whose magnitude increases with the critical ratio $L_2\varepsilon/\gamma$. This empirical evidence supports the theoretical insight that sufficiently strong convexity can compensate for partially ignored decision-dependent effects and still yield high-quality

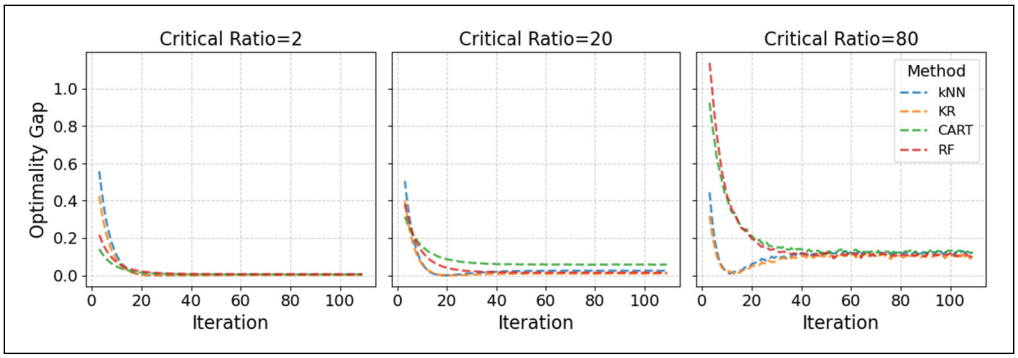


Figure 2. Convergence of contextual gradient descent (CGD) algorithm under different critical ratios.

solutions. Since the loss function is known, quantities such as L_2 and γ can often be computed directly, while the decision-dependence coefficient ε can be estimated from data. Together, they provide an estimate of the critical ratio, which helps assess the expected performance of the algorithm.

5.2.2 Sample efficiency. This section evaluates the sample efficiency of the CGD algorithm by examining its performance under different sample sizes. To ensure that the stable point remains sufficiently close to the optimum, we fix the critical ratio at 2. The kNN weighting scheme is adopted with $k = \lceil 0.1\sqrt{N} \rceil$, and the heteroscedastic demand model is used for data generation. For each sample size, 100 independent instances are generated to evaluate the variability of the algorithm's performance.

Figure 4 reports the mean and standard deviation of the optimality gaps achieved by the CGD algorithm under different sample sizes. Each point corresponds to the empirical mean and variance for a given sample size, while the dashed lines illustrate the overall trends. As shown in the figure, both the mean and variance of the optimality gap decrease as the sample size increases, indicating improved solution quality and stability with larger datasets. Notably, the performance becomes relatively stable once the sample size reaches approximately 300, suggesting that the CGD algorithm can achieve strong performance with only moderate data requirements when the critical ratio is small. These observations are consistent with the finite-sample behavior of the contextual gradient discussed in Section 3.3 and the convergence result in Theorem 3. Specifically, as the sample size increases, the estimation error ζ_N decreases, which tightens the upper bound on the distance between the CGD solution and the optimal solution, thereby reducing the resulting optimality gap.

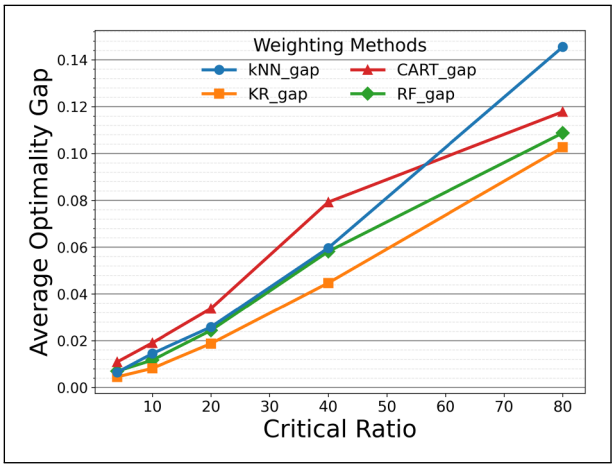


Figure 3. Final average optimality gap against different critical ratio.

5.3 Comparison to other methods

In this section, we first compare the optimization performance of the CGD algorithm with existing solution approaches for the wSAA problem in Section 5.3.1 to demonstrate the effectiveness and computational efficiency of the proposed method. We then compare the CGD algorithm with several representative contextual optimization approaches, including classic ETO, SPO, and end-to-end (E2E) methods, in Section 5.3.2 to highlight its superior performance.

5.3.1 Comparison with directly solving the wSAA model and direct GD. The major advantage of our CGD algorithm is to tackle the computational challenge of directly solving the wSAA formulation under decision-dependent effect. To evaluate the effectiveness of our method, we consider the following two benchmark strategies:

- **Discretization-based optimization** (marked as discretization): In Bertsimas and Kallus (2019), tractability is achieved by discretizing the decision space, an approach that is applicable regardless of the form of the weighting function. In our experiments, both the price p and the order quantity q are uniformly discretized into 100 grid points each over their respective ranges—$[10, 200]$ for price and $[0, 200]$ for quantity. The optimal solution is obtained by evaluating the wSAA objective $\hat{g}(p, q, z)$ over all 10,000 grid combinations and selecting the decision that yields the minimum objective value.

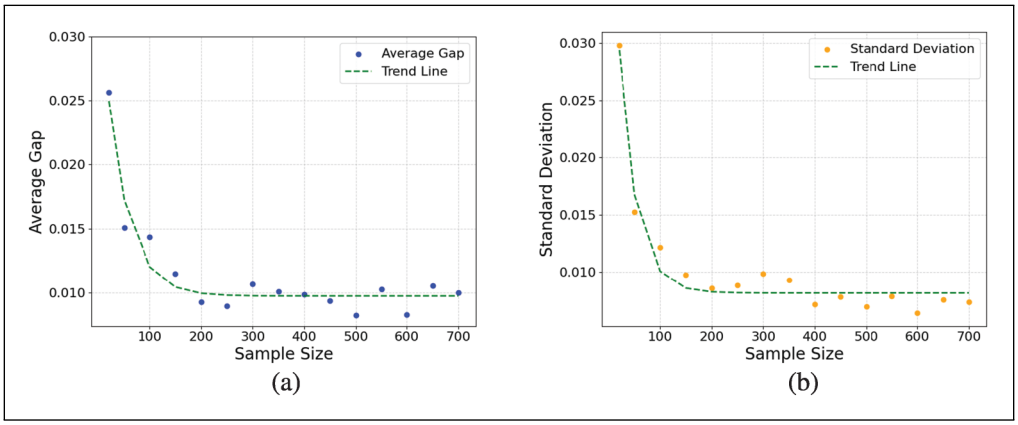


Figure 4. Mean and variance of the optimality gap under different sample sizes. (a) Average gap versus sample size and (b) standard deviation versus sample size.

- **Direct GD on wSAA** (marked as GD): This approach applies standard GD directly to the wSAA objective in (4). As discussed earlier, this method is only applicable when the weighting functions are differentiable with respect to the decision variables, such as those derived from kernel-based methods.

For this experiment, we set the critical ratio to 2, use a dataset of size 1000, and repeat each trial 20 times. The demand is generated using the heteroscedastic model.

We begin by comparing the computational time of these benchmark methods relative to the CGD algorithm. Table 1 presents the average computation times of the discretization method and the CGD algorithm. As shown, CGD significantly reduces computational time compared to the discretization approach, showcasing its efficiency for problems of similar scale.

Directly applying GD to the wSAA only works when the weights are determined by the kernel method. Figure 5(a) compares the average computational time of the CGD algorithm and directly applying GD to solve the wSAA approach across varying data sizes. This result demonstrates that the efficiency benefit of CGD grows with the scale of the problem. Note that wSAA models must be solved for numerous covariate combinations, and therefore the CGD algorithm significantly reduces the total computation time. Moreover, unlike GD, which is limited to differentiable weight functions, CGD applies broadly to a variety of ML-based weighting schemes, making it more versatile.

Table 1. Computation time(s) comparison between CGD and discretization.

Strategies	kNN	KR	CART	RF
Discretization	27.79	25.95	26.99	756.62
CGD	0.41	0.35	0.37	7.59

CGD: contextual gradient descent; kNN: k-nearest neighbor; KR: Kernel regression; CART: classification and regression tree; RF: random forest.

Next, we compare the solution quality of the different methods. Figure 5(b) reports the optimality gaps achieved by the competing approaches under four weighting schemes. Each violin plot represents the distribution of optimality gaps across 20 independent repetitions of the experiment. As shown in the figure, the CGD algorithm consistently achieves solution quality that is comparable to or better than that of the other methods across all weighting functions. This demonstrates that CGD delivers substantial computational savings without sacrificing solution quality. For discretization-based optimization, the achievable optimality gap is fundamentally limited by the grid resolution. For GD, as discussed in Section 4, even when the loss function is strongly convex, the wSAA objective itself need not be convex; consequently, GD may converge to a local optimum rather than the global one. In conclusion, these results highlight that CGD requires considerably less computational effort while achieving comparable or even better solution quality compared to other methods solving the wSAA.

5.3.2 Comparison to other benchmarks. This section compares the CGD algorithm with several additional benchmark methods widely used in the literature to further validate the practical value of the proposed method.

- **ETO):** Following Deng and Sen (2022), this method first fits a demand prediction model using historical observations, where the model class is automatically selected by PyCaret from a pool of candidate regression algorithms. Demand uncertainty is then represented through empirical residuals, that is, $D(p, z) = \hat{\mu}(p, z) + \hat{\varepsilon}$. The optimal price p and order quantity q are subsequently obtained by maximizing the estimated profit under the fitted demand model.
- **Discrete SPO:** Inspired by Elmachtoub and Grigas (2022), this method trains predictive models by directly minimizing downstream decision loss. To preserve linearity, the feasible price space is discretized into a finite

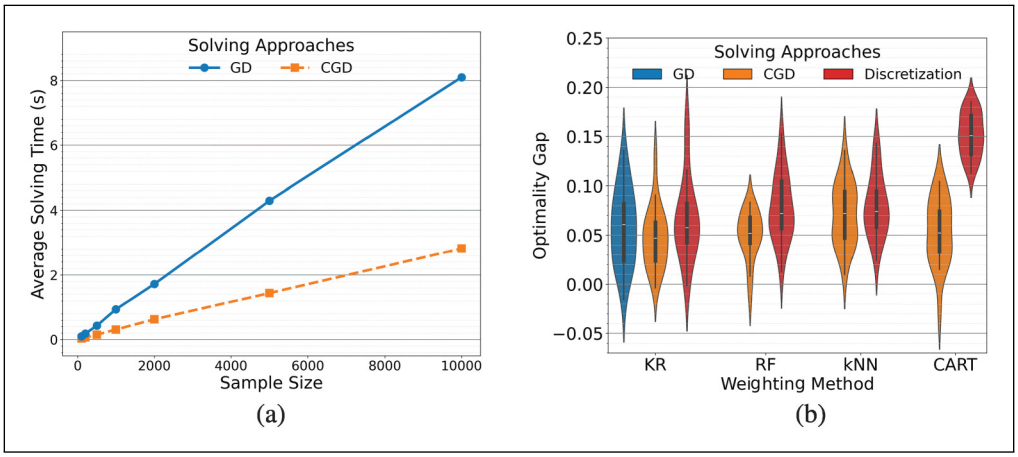


Figure 5. Overall comparison of methods. (a) Computation time of CGD and GD and (b) optimality gap of different solution methods. GD: gradient descent; CGD: contextual GD.

action set, and a predictive model is trained to minimize the loss induced by selecting suboptimal actions.

- **E2E optimization:** Inspired by Ban and Rudin (2018), this method first learns a linear quantity rule $q(x, p)$ from historical data by minimizing empirical operational loss. Given a new context, the price is selected through a discrete search, where the objective value of each candidate price is evaluated using a Gaussian kernel-based wSAA estimator with bandwidth $h = 1$.

We conducted 20 experiments for each of the four methods under both homoscedastic and heteroscedastic demand models, using a training set of size 1000 and a test set of size 50. In these experiments, we set $\varepsilon = \varepsilon_1 = 0.02$ and $\varepsilon_2 = 0.1$ in the homoscedastic and heteroscedastic demand models; all other parameters are consistent with previous sections. Figure 6 compares the performance of the competing methods under two demand models. Figure 6(a) reports the distribution of optimality gaps under the homoscedastic setting using a log scale for clearer visualization. As shown, CGD and the ETO approach achieve comparable performance, and both substantially outperform the other methods. Figure 6(b) presents the results under heteroscedastic demand. In this setting, the performance of ETO deteriorates significantly because its underlying structural assumptions are violated. By contrast, both SPO and E2E are much less sensitive to this misspecification and clearly outperform ETO, showing that end-to-end baselines can provide additional robustness when the data-generating process becomes more complex. Although the CGD algorithm also experiences some performance degradation, it consistently maintains an optimality gap below 5%. This robustness highlights the advantage of CGD's nonparametric nature, which allows it to flexibly capture more complex, decision-dependent relationships.

In summary, CGD performs comparably to ETO when the structural assumptions underlying ETO hold, and significantly outperforms ETO when those assumptions are violated. Moreover, while SPO and E2E methods provide more robust alternatives than ETO under heteroscedastic demand, they still remain consistently inferior to CGD in terms of solution quality. In practice, the interactions among decisions, covariates, and stochastic outcomes are too complex to specify parametrically, and therefore, the nonparametric nature of CGD provides superior robustness and consistently higher solution quality.

6 Case study

In this section, we conduct a real-world case study to evaluate the empirical performance of the proposed CGD algorithm when the theoretical assumptions may not strictly hold. We study a pricing and inventory problem for used-vehicle sales using a dataset published on Kaggle.[1] The problem naturally fits the decision-dependent CSO framework: the selling price directly influences demand, while optimal pricing decisions depend on contextual information such as market conditions (e.g., Manheim Market Report values and regional trends), vehicle attributes (brand, model, trim, and type), and dynamic characteristics (mileage and condition rating). Details of the data preprocessing and modeling procedures are provided in EC.6.1 and EC.6.2. The dataset is split into a training set (80%) and a test set (20%). The training set is used to construct the contextual gradients. Among the candidate weighting methods, the RF weighting scheme (Example 4) achieves the best predictive performance in cross-validation and is therefore adopted in the experiments. For each covariate realization in the test set, the CGD algorithm is applied to obtain the corresponding price and order quantity decisions. To evaluate solution quality, we additionally train 24 ML models to capture the relationship between sales volume, selling price, and covariates.[2] Among them, the Extra Trees model achieves the best predictive performance, with an overall R^2 of 0.87. Therefore, for each test instance and pricing decision, the predicted sales generated by this model are used to compute the corresponding newsvendor profit.

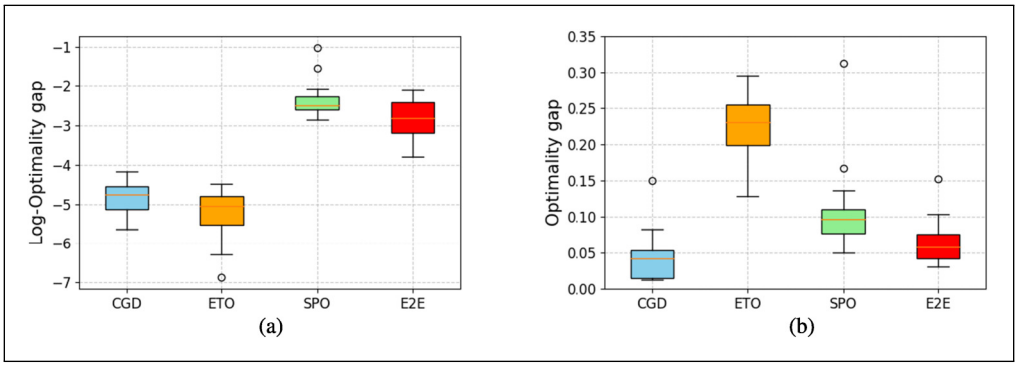


Figure 6. Performance comparison of different methods. (a) Homoscedastic demand model and (b) heteroscedastic demand model.

Figure 7 reports the relative revenue improvement achieved by the CGD-derived decisions compared with the original decisions observed in the dataset. The red horizontal line corresponds to a ratio of one, while the green line indicates the average ratio. As shown in the figure, the CGD algorithm delivers substantial and robust revenue improvements: the average revenue gain, evaluated under the Extra Trees demand proxy, is 31%, and only 3.7% of the test instances experience a slight revenue decrease. These results demonstrate the practical value of the proposed CGD algorithm for real-world pricing and inventory optimization problems.

7 Conclusion and future directions

In this paper, we propose the CGD algorithm to solve CSO problems with decision-dependent uncertainty. Specifically, we introduce the *contextual gradient* as a wSAA of the expected gradient of the loss function, and use it as a tractable and easy-to-compute proxy for the descent direction. This computational advantage comes at the cost of ignoring part of the decision-dependent effect; consequently, the contextual gradient constitutes a biased approximation of the true gradient. Nevertheless, our analysis shows that the iterates generated by the CGD algorithm converge to a neighborhood of the optimal solution when the loss function is sufficiently strongly convex relative to the magnitude of the decision-dependent effect. This condition is characterized by $\gamma \geq \varepsilon L_1^c$, where γ denotes the strong convexity parameter, ε measures the sensitivity of the distribution with respect to the decision, and L_1^c bounds the variation induced by the decision-dependent component. Moreover, the size of this neighborhood shrinks linearly with the ratio ε/γ. These results imply that CGD is particularly well suited for problems in which the loss function exhibits sufficient curvature relative to decision-dependent effects—a condition that commonly arises in practice due to regularization, adjustment costs, or risk considerations. More broadly, a key insight from our analysis is that strong convexity can partially compensate for the unmodeled decision-dependent effects, thereby ensuring reliable optimization performance despite the biased descent direction.

There are multiple promising directions for future research. One is the development of alternative algorithms that integrate the contextual gradient, such as adaptations within proximal GD or SGD frameworks. Another direction is the extension of CGD to settings involving multistage decisions or time-varying distributions. These extensions would further broaden the applicability and impact of contextual optimization under decision dependency.

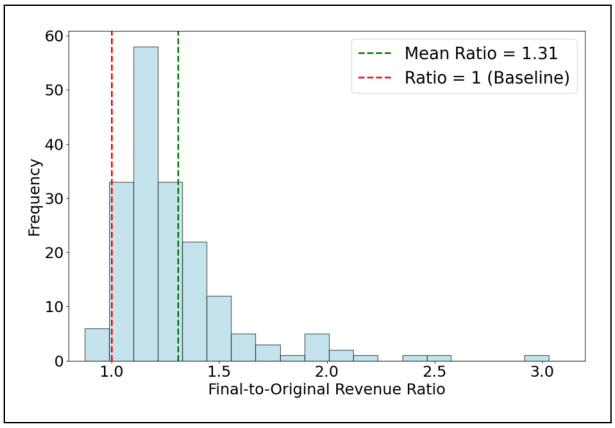


Figure 7. Histogram of final to original revenue ratios.

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Supplemental material

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Notes

1. https://www.kaggle.com/datasets/syedanwarafridi/vehicle-sales-data
2. Detailed results are reported in Table EC.4 in the EC.

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